

Solution for Problem 1.

claim! There cannot exist a pair of positive integers (m, n) such that $2^m + 3^m + 5^m = n!$

Proof! The minimum value for LHS is when $m=1$, $2^1 + 3^1 + 5^1 = 10$, therefore the value of $n! \neq 10$ ($n!0$ cannot be expressed in a factorial form) and $3! = 6$, $4! = 24$ therefore $n > 3$.

When n is greater than 3, $n!$ must always contain prime factors of 2 and 3, which means in order for LHS, $2^m + 3^m + 5^m$, to equal RHS, $n!$, LHS must also have prime factors 2 and 3.

Assuming for the sake of contradiction that $2^m + 3^m + 5^m$ can equal $n!$ for some positive integers (m, n) then $2^m + 3^m + 5^m$ must be divisible by 2 and 3.

We can prove that $2^m + 3^m + 5^m$ is always divisible by 2 as 2^m contains a prime factor of 2 which means it must be even, 3^m doesn't contain a prime factor of 2 which means it must be odd, and 5^m doesn't contain a prime factor of 2 which means it must also be odd. ~~An odd even number + an odd number~~ Even + odd + odd must be even as the two odds add up together to form an even and the even + even must be even. Therefore $2^m + 3^m + 5^m$ always have a prime factor of 2.

However, $2^m + 3^m + 5^m$ cannot be divisible by 3 as $(2^m + 3^m + 5^m) \pmod{3} \equiv (0 + 5^m + 2^m) \pmod{3}$ ($5^m = (3+2)^m$ which can be expressed as $3N_1 + 2^m$ for some positive integer N_1) Therefore $(2^m + 3^m + 5^m) \pmod{3} \equiv (0 + 5^m + 2^m) \pmod{3} \equiv (3(N_1) + 2^m + 2^m) \pmod{3} \equiv (2^{m+1}) \pmod{3}$ (for any positive integer of m)
Since 2^{m+1} is in its prime only have 2 as its prime factor and doesn't have 3, $(2^{m+1}) \pmod{3} \not\equiv (0) \pmod{3}$, therefore $(2^m + 3^m + 5^m) \pmod{3} \not\equiv (0) \pmod{3}$ ~~this means~~ and that $2^m + 3^m + 5^m$ doesn't have a prime factor of 3. This contradicts the initial assumption of $2^m + 3^m + 5^m$ being able to equal $n!$ for positive integers (m, n) as then LHS must have 3 as a prime factor, and therefore proving that there cannot exist a pair of positive integers (m, n) such that $2^m + 3^m + 5^m = n!$